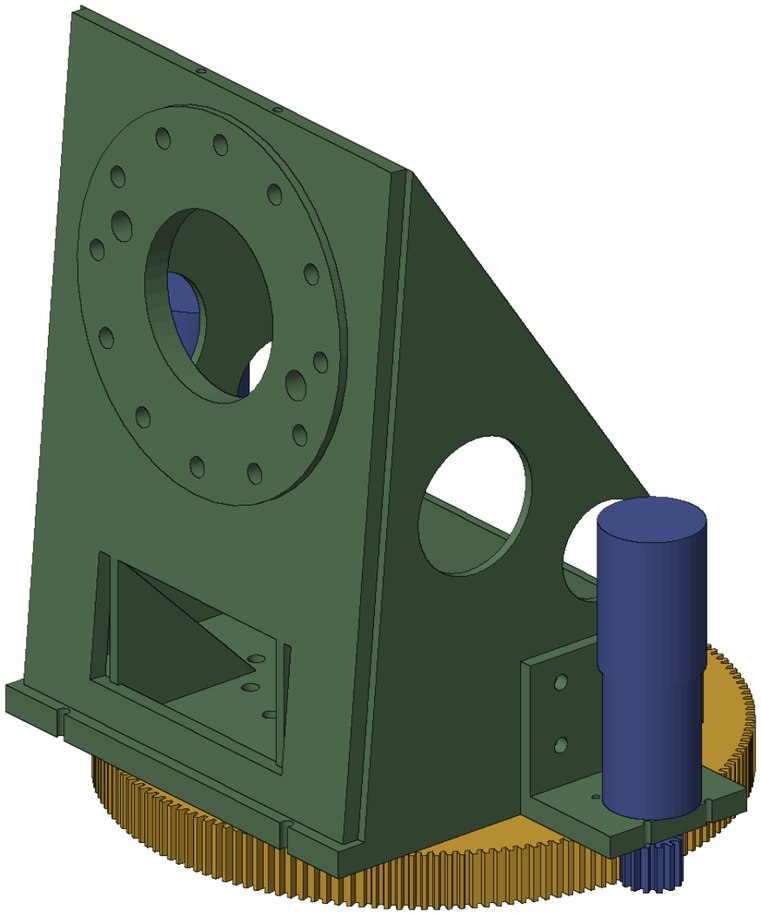
6 Assembly sequence

Workshop steps 1–11, site steps 12–13

6 Assembly sequence

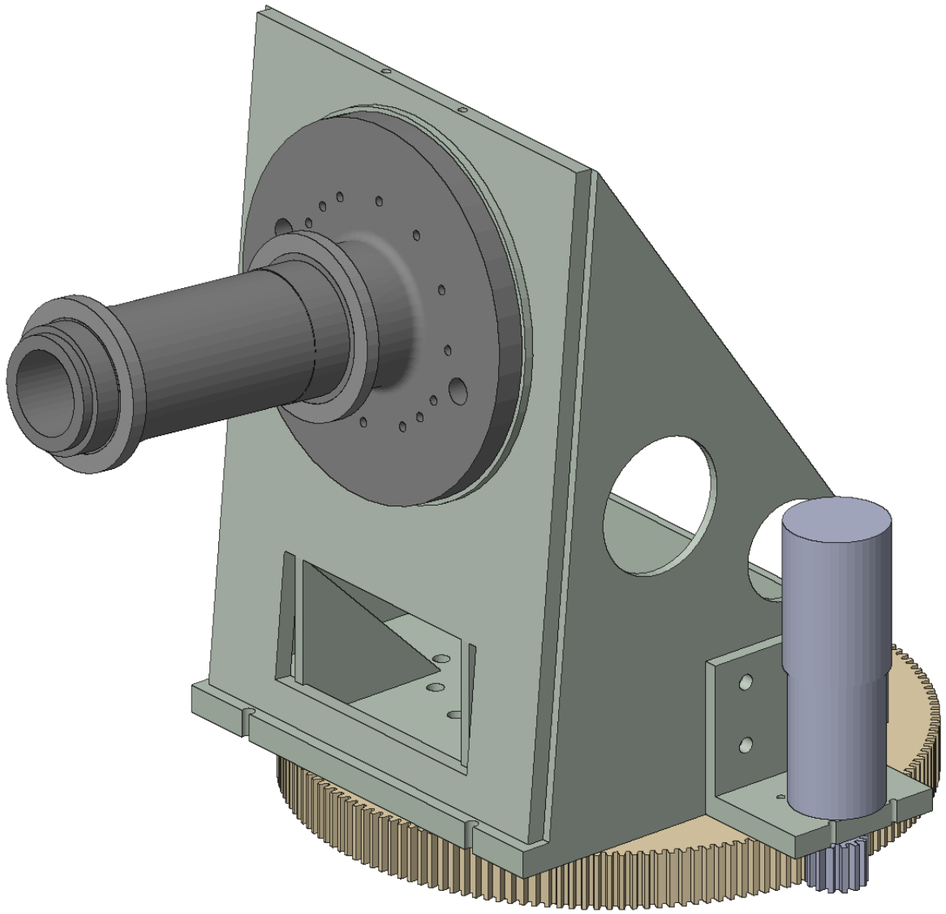
Step 1 Bedplate NP8-300 on the stand; yaw slewing ring Dpw 410, gearmotors and their brackets



- Set the NP8-300 bedplate upside-down on the assembly stand; check the yaw flange flatness (0.05) and the front-face angle (5.00° ± 0.05°) and record them.
- Fit the Dpw 410 slewing ring inner ring: 16 × M16 HV on PCD 350, cross pattern, 280 Nm (soft gearing mark towards the lock side).
- Fit the gearmotor brackets NP1-420 and the two gearmotors; set the backlash 0.15–0.30 mm at the soft point with the bracket shims.
- Check: ring turns freely by hand with both motor brakes released; record the turning torque.

Parts (new in colour, earlier steps light): NP8-300, NM8-911-I, NM8-911-O, NM8-912-L, NM8-912-R, NP1-420-L, NP1-420-R. Torques: NM8-900.

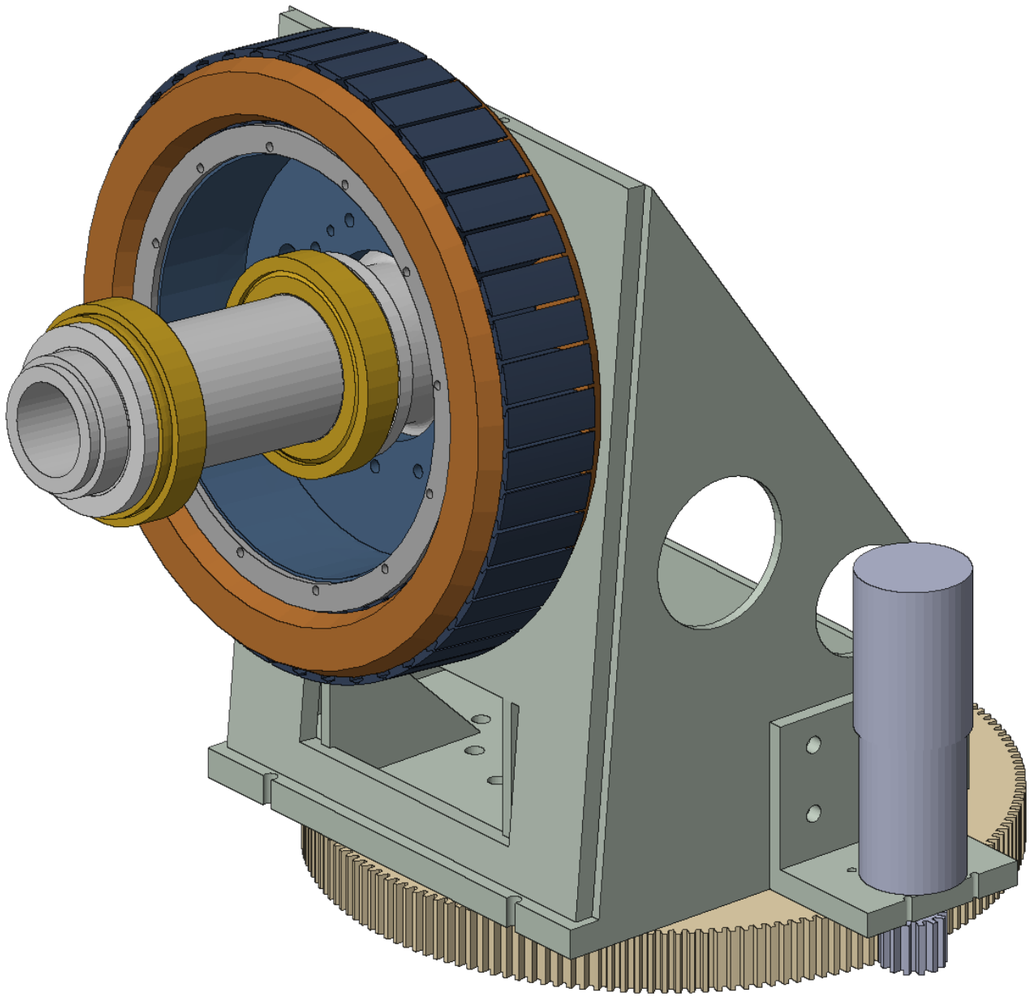
Step 2 Spindle Rev D with front spacer and rear seal sleeve bolted to the bedplate front plate



- Clean the spindle flange and pad; fit the Ø200 spigot; 12 × M20 10.9 from inside, 540 Nm in three passes.
- Fit the front spacer NP1-101 and rear seal sleeve NP1-102 (heat to 120 °C); cables through the two Ø28 holes before the stator goes on.
- Check: flange face square to the yaw axis (5° tilt) within 0.05/400. Spindle Rev D is 30 mm longer: check the length 466 ± 0.1 before fitting.

Parts (new in colour, earlier steps light): NP1-100D, NP1-101, NP1-102. Torques: NM8-900.

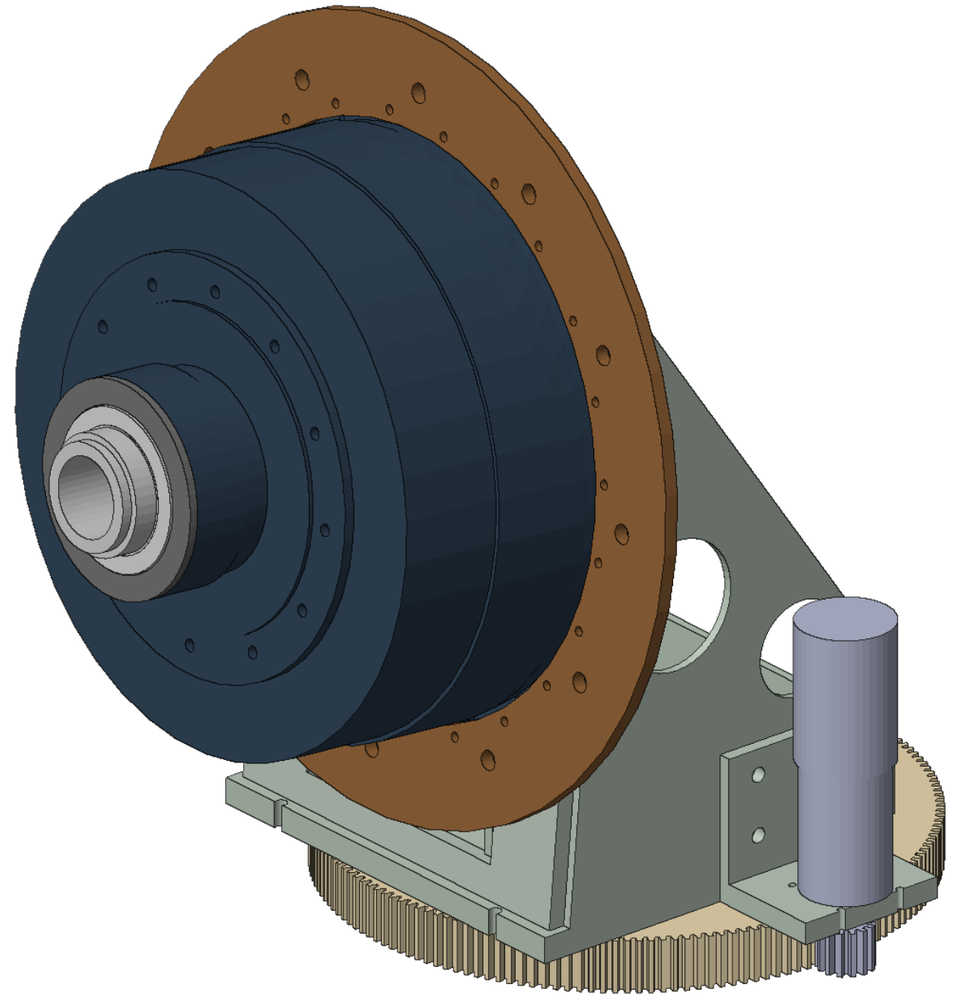
Step 3 Stator Rev D: carrier, wound tooth segments (104 mm), clamp ring; bearing cones on the spindle



- Assemble the stator Rev D: 45 wound tooth segments (104 mm stacks) slid into the carrier dovetails, clamp ring NP1-202 (doc 18 instructions, Rev C with the Rev D lengths).
- Insulation 1 kV, winding resistance balance ≤ 2 %, surge test. Carrier to spindle: 12 × M12, 110 Nm (thermal paste).
- Heat the bearing cones (110 °C) and fit them; grease 60 % fill.

Parts (new in colour, earlier steps light): NP1-200D, NP1-201D, NP1-203D, NP1-202, NM8-913-F, NM8-913-R. Torques: NM8-900.

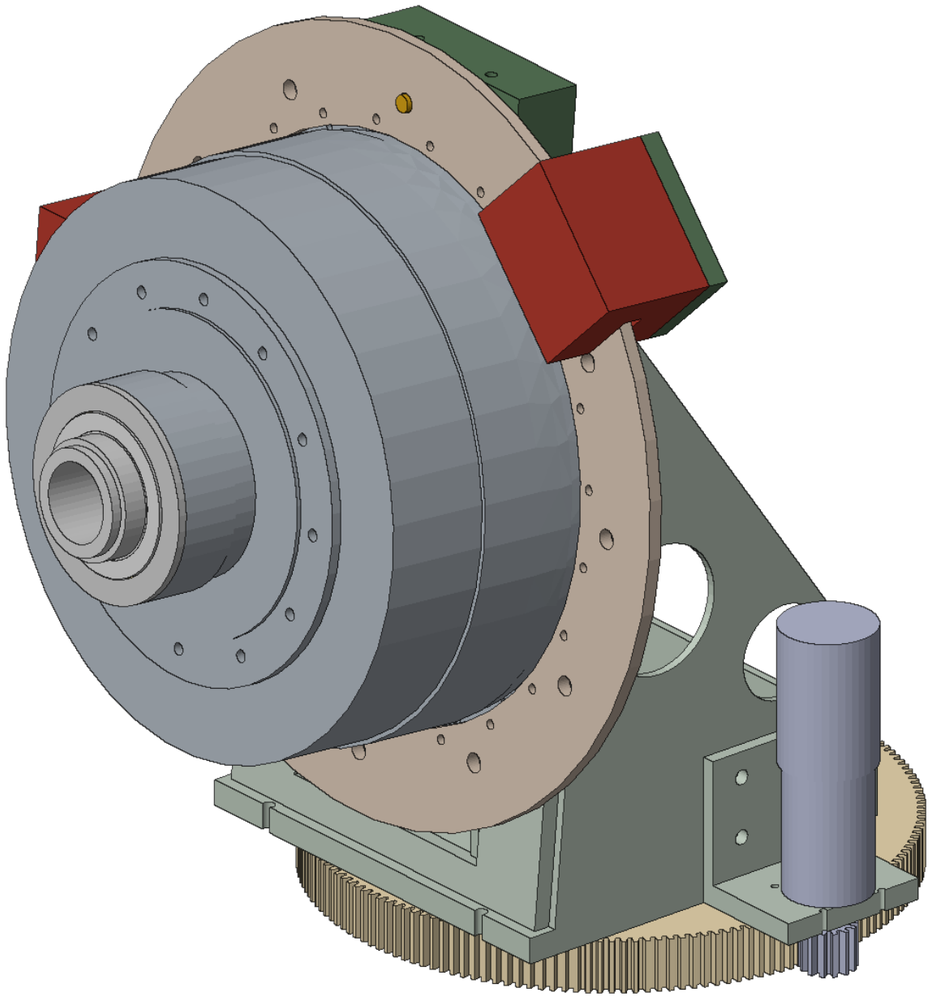
Step 4 Rotor drum Rev D with 7 magnet rows lowered over the stator on the guide tool; disc, seal covers, rear cover



- Magnet drum: use the drum guide NP1-T01 (lengthened 30 mm) and the magnet safety rules of doc 18 (no steel tools within 1 m, two people). 280 blocks in 7 rows, skew steps ±0.4°.
- Lower the drum on the guide screw; set the bearing preload with the KMT 27 locknut (rolling torque 6–8 Nm).
- Fit the disc (24 bolts, NM8-900), seal covers and the Rev C rear cover with fan blades. Remove the guide.
- Check: air gap ≥ 1.4 mm at 8 positions, drum run-out ≤ 0.2 mm, back-EMF at 60 rpm vs generator_m8 (+3.4 % on Rev C).

Parts (new in colour, earlier steps light): NP1-210D, NP1-220F, NP1-230, NP1-211-F, NP1-211-R, NP1-212C. Torques: NM8-900.

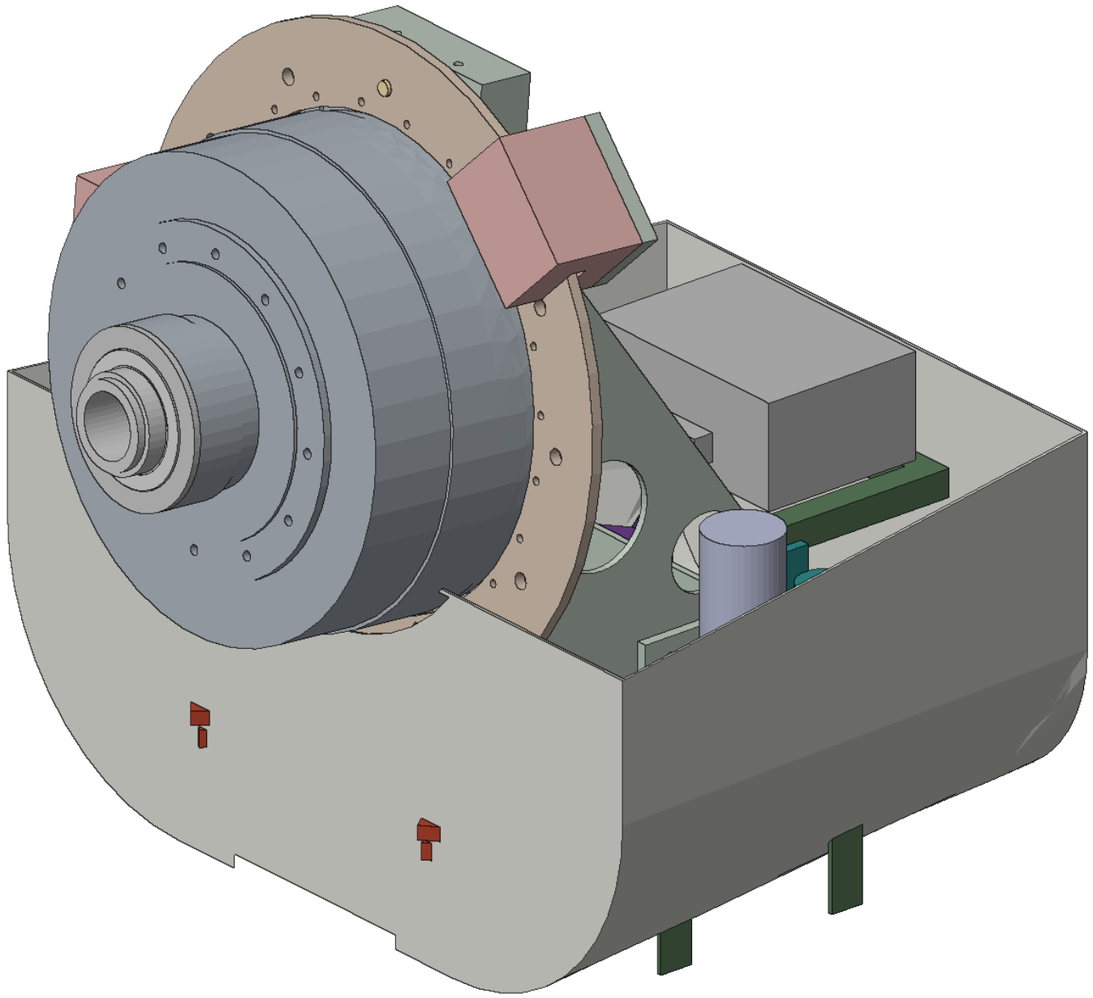
Step 5 Three brake calipers on their brackets, lock pin holder with overspeed sensors



- Fit the two side caliper brackets NP1-310 and the third bracket NM8-311 (4 × M16 through the front plate). Fit the three calipers; set pad gaps; check release on 24 V.
- Fit the lock pin holder with the two overspeed sensors; test the lock pin in all 12 disc holes.
- Check: holding torque test 14.4 kNm (torque arm on the drum), each caliper alone ≥ 4.8 kNm; sensor pulses.

Parts (new in colour, earlier steps light): NP1-310, NM8-915, NM8-311, NM8-915-T, NP1-320, NP1-321, NM8-914. Torques: NM8-900.

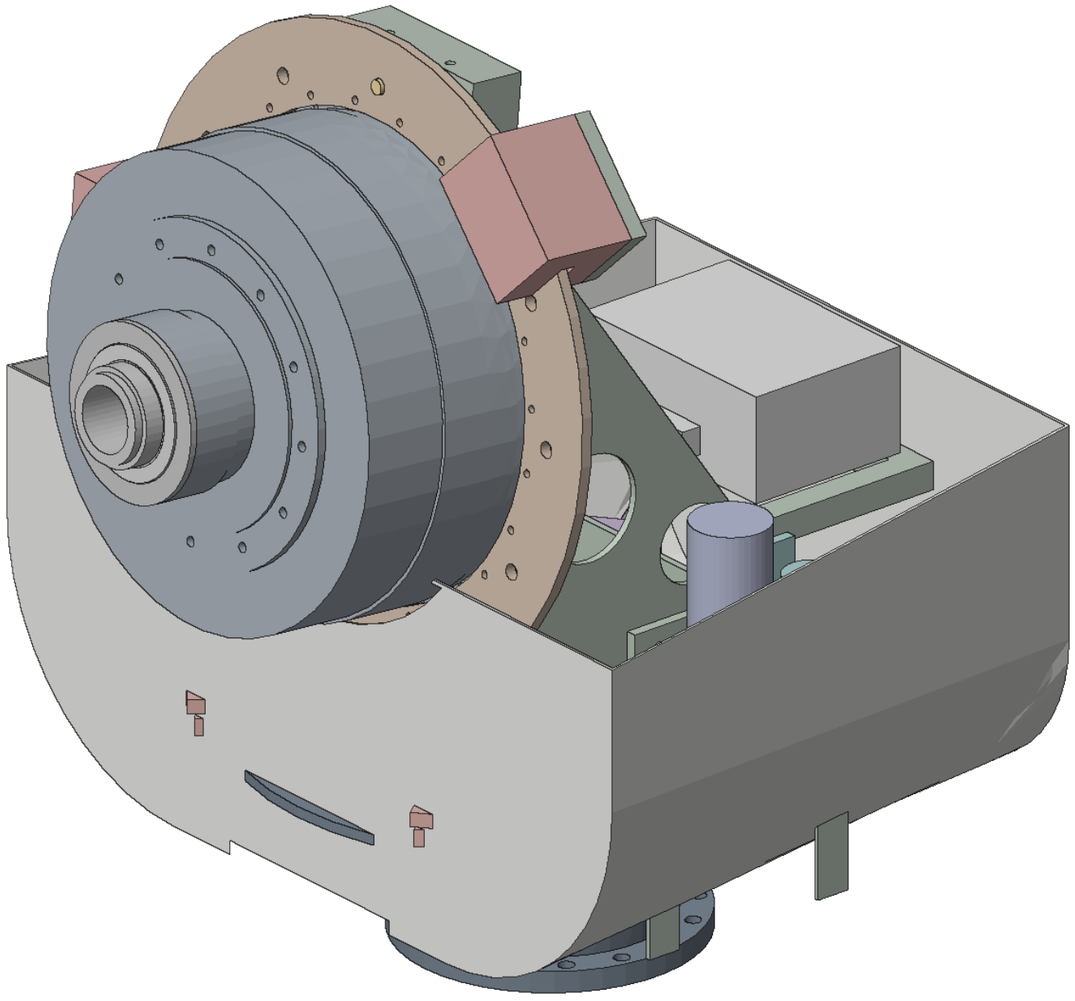
Step 6 Yaw pad units, lock, encoder and cable hanger; cover frame, lower cover tray and junction box



- Fit the pad units NY-410 (radius +25), yaw lock NY-430, encoder unit NY-440 and the cable hanger NY-460.
- Fit the cover arms, the lower GRP tray, the rear rail and the junction box; wire W1, W4, W5, W6.
- Check: pad clamping force, encoder zero at yaw 0°, cam switch at ±2.5 turns.

Parts (new in colour, earlier steps light): NY-410-L, NY-410-R, NY-430, NY-440, NY-460, NM8-710, NM8-700-1, NM8-730. Torques: NM8-900.

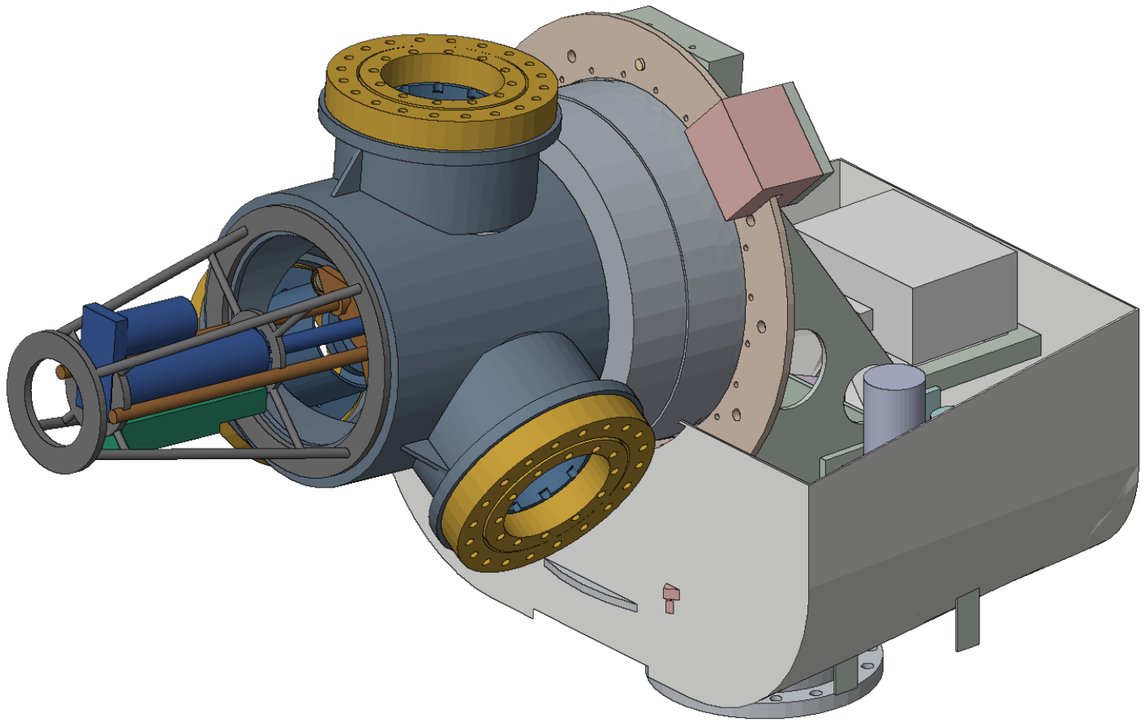
Step 7 Tower-top adapter on the yaw outer ring (nacelle now complete for transport)



- Lower the nacelle onto the tower-top adapter NY-400 (outer ring 16 × M16 PCD 490, 280 Nm). The nacelle is a transport unit with lifting points on the bedplate.
- Check: yaw turns 360° with the drives; pads slip at the doc 25 torque (scaled to the Dpw 410 ring).

Parts (new in colour, earlier steps light): NY-400. Torques: NM8-900.

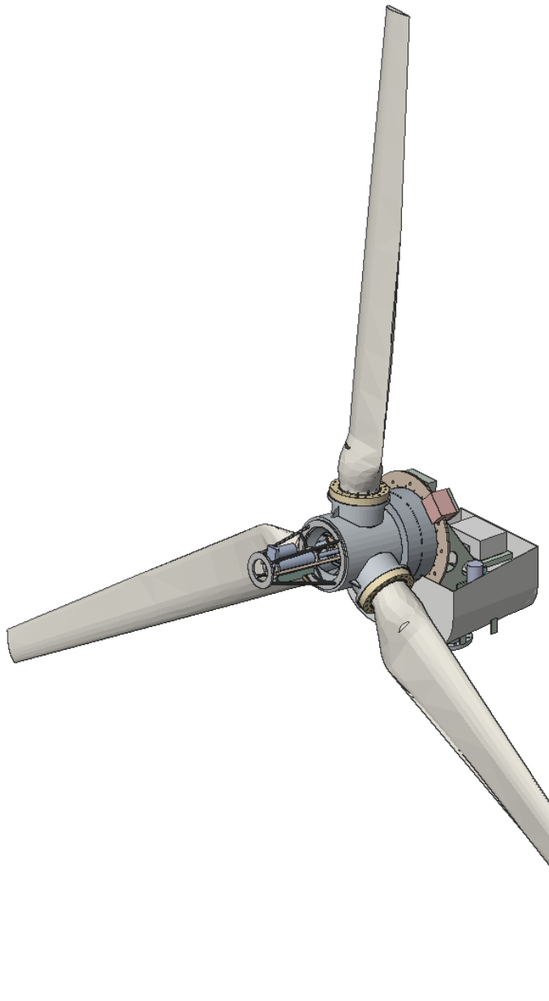
Step 8 Pitch hub pre-assembled on the bench (weldment, pitch bearings, cranks, links, yoke, cylinder, supercaps, nose frame), then on the drum flange (12 × M16 10.9)



- Bench: weld-checked hub NM8-200; fit the three pitch bearings (20 × M16 each, 280 Nm), cranks r140 with pins, links, yoke and guide rods, the cylinder and motor, supercap module and drive, nose frame.
- Run the pitch system on the bench from −2° to 92° and back (limit switches, end stops, supercap feather test, time ≤ 4 s).
- Fit the hub on the drum spigot (Ø360), 12 × M16 10.9, 280 Nm in 3 passes; mark blade 1 at the lock-pin azimuth; connect the slip ring.

Parts (new in colour, earlier steps light): NM8-200, NM8-250, NM8-260, NM8-261, NM8-262, NM8-270, NM8-280, NM8-290. Torques: NM8-900.

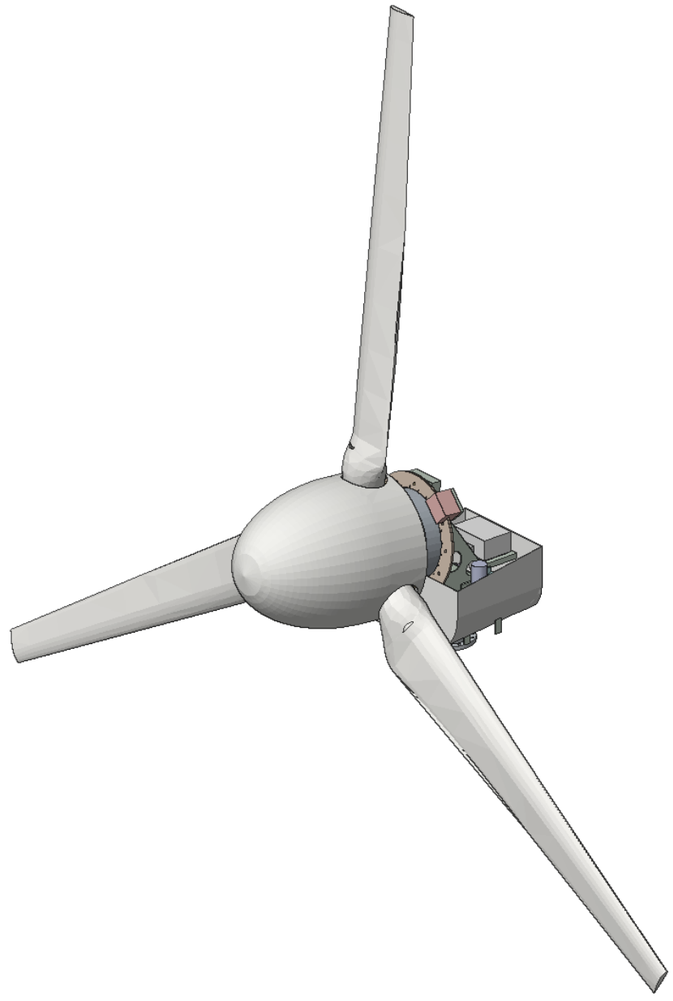
Step 9 Blades on the pitch bearings (12 × M16 each), pitch zero set with the cylinder at its −2.0° end stop



- Cylinder at the −2° end stop. Mount B1 on bearing 1, B2, B3 by their static-moment record: 12 × M16 10.9, 280 Nm star pattern.
- Check with the inclinometer at r 2 000 that all three blades read −2.0° ± 0.1° (linkage adjusted at the rod ends); tip track ±15 mm.

Parts (new in colour, earlier steps light): NM8-100-1, NM8-100-2, NM8-100-3, NM8-105. Torques: NM8-900.

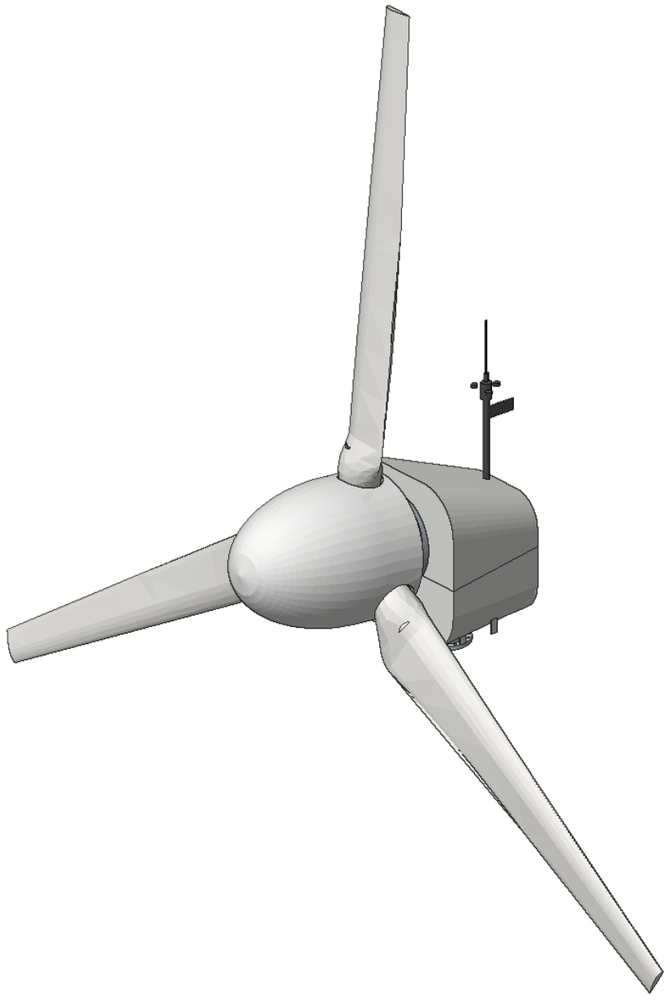
Step 10 Spinner with blade boots



- Fit the three spinner segments with the blade boots (9 brackets, A4 M8, 12 Nm). Check 10 mm clearance to all pitch parts over the full stroke.

Parts (new in colour, earlier steps light): NM8-210. Torques: NM8-900.

Step 11 Upper cover lid and met mast



- Fit the upper cover lid and the met mast; bond the air terminal to the bedplate (50 mm² Cu).
- Factory test: pitch, brake, yaw, sensors; record insulation and EMF.

Parts (new in colour, earlier steps light): NM8-700-2, NM8-720. Torques: NM8-900.

Step 12 On site: footing with anchor cage, hinge brackets, tower lying on the hinge; head bolted to the tower top



- Site: excavate to 1.6 m, blinding, bottom mesh, anchor cage NM8-510 with the template ring (± 3 mm, level ± 1 mm), hinge anchors, ducts, earthing ring.
- Pour C30/37 XF4, cure ≥ 7 days. Fit the hinge brackets NM8-410 and levelling nuts; lay the tower on trestles on the hinge pin.
- Bolt the head to the tower top (16 × M20 HV, 540 Nm) on the ground; pull W2/W5; rotor locked with B1 up, blades feathered.

Parts (new in colour, earlier steps light): NM8-500, NM8-510, NM8-410, NM8-400. Torques: NM8-900.

Step 13 Tilt-up with the gin pole and winch, base nuts torqued; cabinet, resistor box, cabling, hand-hole cover



- Gin pole and rigging; winch line pull ≈ 66 kN (use a 2-part or 4-part line); wind below 8 m/s; nobody in the fall zone.
- Land on the levelling nuts, fit top nuts, grout 30 mm; after 7 days tension the anchors (NM8-900). Remove the gin pole.
- Cabinet, resistor box, cabling W3, earthing ≤ 10 Ω, hand-hole cover. Commissioning checklist §7.

Parts (new in colour, earlier steps light): NM8-420, NM8-620, NM8-600, NM8-800, NM8-401. Torques: NM8-900.

7 Erection, commissioning and inspection

Before power-up

- All torque stripes present; anchor, adapter, hub, pitch bearing and blade bolt records signed (NM8-900).
- Tower plumb within 0.3 % in two directions; grout cured; hand-hole cover fitted.
- Earthing ≤ 10 Ω; continuity air terminal → bedplate → pad brush → tower → earth ring ≤ 0.2 Ω; hub bonded across the slip ring.
- Insulation: generator ≥ 100 MΩ at 1 kV; cables ≥ 100 MΩ. Pitch supercap charged, heater on below 0 °C.
- Rotor lock pin out; yaw lock retracted; cover lid closed; nobody inside 24 m.

Functional tests (wind below 8 m/s)

- Safety chain: each trip input (overspeed × 2, E-stop, 24 V loss, vibration, converter fault, pitch fault) feathers the blades on supercap power and drops the brake after the delay; record the stop time.
- Feather on the supercap alone (mains off) from rated speed; brake alone from 100 rpm with the blades at −2°.
- 8 m controller settings loaded: speed loop 6 rad/s, normal pitch rate 20°/s, overspeed trip 165 rpm (18 m: 4 rad/s, 15°/s, 155 rpm). Trip checked by lowering the set point (test mode).
- Yaw ±360° with the cam stop at ±2.5 turns; exclusion zone 48–61 rpm set in the controller (passed under torque).
- Rotor balance at 120 rpm: nacelle 1P vibration, trial mass to ≤ 0.2 kg·m; pitch angle check (all three within 0.3°) with a camera at the tips.
- Tower frequency (tap test): f1 must be ≥ 2.65 Hz; if lower, widen the exclusion zone.
- Power curve: 48 h supervised operation; pitch regulation above 10 m/s; storm feather test on the next windy day.

Inspection plan

Interval	Inspection
50 h	Re-torque blade root, pitch bearing, hub, adapter and anchor bolts; pitch zero check
3 months	Blades (LE erosion, root cracks), brake pads, pitch linkage play, cover fixings
Yearly	Bolts 10 % sample; pitch bearing grease; cylinder and rod ends; supercap capacity test; slewing ring and yaw gear grease; main bearing grease; earthing; safety chain; brake torque
5 years	Lower the tower: full blade inspection, main bearing play, magnet sleeve, cable loop, pitch bearing play
After a storm > 50 m/s or an ice event	Visual check, pitch and yaw logs, brake slip log

Lowering the tower for service

- Stop, feather, yaw to the hinge direction, lock the rotor with B1 up, lock the yaw; rig the gin pole and winch; slacken the base nuts on the hinge side last.
- Lower to the trestle: nacelle, hub and pitch system are at working height.

Records to keep with the turbine

- Blade set static moments; magnet certificates; weld and UT records (hub, tower, bedplate, caliper bracket); torque records; commissioning results; f1 measured.

8 Open items before the prototype release

- Martin's decisions D1–D5 (doc 35): this book uses the recommended defaults.
- FE checks: hub neck-to-shell welds, NP8-300 ring seat and front plate with the third caliper bracket, spindle Rev D flange fillet.
- Supplier drawings replace the envelopes (pitch bearings, cylinder, slewing ring Dpw 410, gearmotors, calipers, main bearings); then rerun clash8.py.
- Blade first-article static test to SLM H × 1.35, insert pull-out, mould acceptance.
- Pitch system bench test (feather on supercap at −30 °C).
- Site investigation: soil class, frost depth, earthing, obstacle survey at 8 m; snow-drift height under the rotor.
- Tower frequency on site (≥ 2.65 Hz) and the fatigue at the base: measured strain on the prototype for one winter.
- Energy at 8 m: the 8 m tower is for open coast, highland and test sites; sheltered sites need the 18 m tower.
- All drawings NOT CHECKED: independent check of every sheet before parts are ordered.